

Session 15 Quiz - Computer Vision

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 53600]

The branch of AI focused on processing and interpreting visual information from images and videos is called _____ Vision.

- A. Computer
- B. Natural
- C. Synthetic
- D. Digital

Correct Answer: A

Explanation: Computer Vision (CV) is the AI subfield focused on visual data. It parallels Natural Language Processing - where NLP handles text and language, CV handles images and video. "Synthetic" and "digital" are general terms without specific AI meaning.

Q2 [Type: MCQ | Difficulty: Easy | QID : 53601]

A _____ is the smallest unit of a digital image - a single dot of colour, defined by its red, green, and blue values.

- A. Feature
- B. Pattern
- C. Pixel
- D. Prediction

Correct Answer: C

Explanation: Pixels are the raw building blocks of every image. A pure red pixel is (255, 0, 0); pure white is (255, 255, 255); pure black is (0, 0, 0). CV systems start by reading these numbers, not the image you "see."

Q3 [Type: MCQ | Difficulty: Medium | QID : 53602]

_____ is the Computer Vision task that reads text from images - for example, converting a photo of a printed sign into editable text on the clipboard.

- A. Image Classification
- B. Object Detection
- C. Facial Recognition
- D. Optical Character Recognition

Correct Answer: D

Explanation: OCR is the task specifically for turning pixels of letters into digital text. Tools like Google Lens use OCR to extract text from photos so you can search, copy, or translate it. The other tasks handle different visual problems.

Q4 [Type: MCQ | Difficulty: Easy | QID : 53603]

Computer Vision systems truly see and understand images the same way a human eye and brain do.

- A. True
- B. False

Correct Answer: B (False)

Explanation: CV finds mathematical patterns in pixels and guesses what those patterns represent - it does not "see" in the human sense. The AI does not know what "stop" means; it has learned that certain pixel patterns appear together with the label "stop sign" in its training data. CV is pattern matching at scale - which is exactly why it can fail on lighting, angles, and occlusion in ways humans never would.

Q5 [Type: MCQ | Difficulty: Medium | QID : 53604]

If a Computer Vision system works reliably during the daytime, it will work just as well at night with the same accuracy.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Lighting is one of CV's named challenges. The same stop sign produces very different pixel values for the same features (colour, edges, brightness) in bright sun versus darkness. A system trained mostly on daytime images can miss the same object entirely after dark. Engineers explicitly test CV models in many lighting conditions for this reason.

Q6 [Type: MCQ | Difficulty: Medium | QID : 53605]

The more pixels a digital image contains, the more data the AI has to work with for finding features and patterns.

- A. True
- B. False

Correct Answer: A (True)

Explanation: A higher-resolution image gives CV more pixels to extract features from, which generally helps with sharper edges, finer texture, and small distant objects. This is directly connected to the "Scale" challenge: a pedestrian far away occupies only a few pixels, leaving the AI very little to work with. The mosaic analogy says this explicitly - the more tiles, the sharper the picture.

Q7 [Type: MCQ | Difficulty: Easy | QID : 53606]

What is the correct order of the three stages of the Computer Vision framework?

- A. Act → See → Process
- B. Process → Act → See
- C. See → Process → Act
- D. Predict → Process → Output

Correct Answer: C

Explanation: See (capture pixels) → Process (Features → Patterns → Prediction) → Act (do something with the result). This matches the Sense → Think → Act framework introduced earlier in the course.

Q8 [Type: MCQ | Difficulty: Easy | QID : 53607]

In the Process stage of Computer Vision, which of the following is the AI's "guess" about what an image contains - often given with a confidence score?

- A. Pixel
- B. Feature
- C. Pattern
- D. Prediction

Correct Answer: D

Explanation: Pixels are the raw input. Features are clues extracted from pixels. Patterns are combinations of features. Prediction is the final guess (e.g., "95% confident - this is a stop sign") - and Action is what the system does next.

Q9 [Type: MCQ | Difficulty: Medium | QID : 53608]

Which Computer Vision task finds WHERE objects are in an image by drawing bounding boxes around them?

- A. Image classification
- B. OCR
- C. Facial recognition
- D. Object detection

Correct Answer: D

Explanation: Object detection locates objects in the image and outlines them with bounding boxes. This is what a self-driving car uses to flag a car at one location, a pedestrian at another, and a traffic light at a third - all in the same scene.

Q10 [Type: MCQ | Difficulty: Medium | QID : 53609]

Which Computer Vision task LABELS the main object in an image - for example, looking at a photo and outputting "cat"?

- A. Object detection
- B. OCR
- C. Image classification
- D. Facial recognition

Correct Answer: C

Explanation: Image classification produces a single label for the main object in an image. Object detection finds multiple objects and their locations; facial recognition identifies who a face belongs to; OCR reads text from pixels.

Q11 [Type: MCQ | Difficulty: Medium | QID : 53610]

Which component of the "Process" stage refers to the combinations of features that repeat across many training examples?

- A. Pixels
- B. Patterns
- C. Sensors
- D. Predictions

Correct Answer: B

Explanation: Patterns are established when the AI recognises that specific features, like two eyes and a mouth, consistently appear together in a certain arrangement. Pixels are the raw input, predictions are the final guess, and sensors are part of the See stage (the camera).

Q12 [Type: MCQ | Difficulty: Medium | QID : 53612]

A pixel with RGB values of (0, 0, 0) represents which of the following?

- A. Pure White
- B. Pure Black
- C. Transparent
- D. Neutral Grey

Correct Answer: B

Explanation: The absence of all colour channels (Red, Green, Blue all at zero intensity) results in pure darkness. (255, 255, 255) is pure white - maximum of all three channels. (0, 0, 0) is pure black.

Q13 [Type: MCQ | Difficulty: Medium | QID : 53614]

The material emphasises that Computer Vision is "pattern matching, not true understanding." Which example BEST illustrates this distinction?

- A.** The self-driving car correctly recognises a stop sign, but it does not know what "stop" means - it has learned that certain pixel patterns appear together
- B.** The system uses a confidence score with its prediction
- C.** The system processes the image quickly
- D.** The system outputs its result on a screen

Correct Answer: A

Explanation: The car correctly executes the action not because it conceptually understands "stop," but because it matched the pixel pattern to learned examples. Pattern matching can succeed without comprehension - and it can fail when the pattern shifts (snow, darkness, occlusion). The other options describe surface features that have nothing to do with understanding.

Q14 [Type: MCQ | Difficulty: Hard | QID : 53615]

A self-driving car's pedestrian detection system was trained mostly on images of adults. When the car encounters a small child on the road, it sometimes fails to recognise the child as a pedestrian. Which CV challenge is most clearly at work?

- A.** Lighting - children are often photographed indoors, so the model struggles with outdoor scenes
- B.** Background clutter - children often have toys around them, which distracts the detection system
- C.** Bias - the training data underrepresented children, so the model recognises adults more reliably
- D.** OCR failure - the system cannot read the text on the child's clothing to identify them as a pedestrian

Correct Answer: C

Explanation: Underrepresentation in training data produces a system that performs worse on the missing group - and when the missing group is children at risk on the road, the bias becomes a safety issue, not just an accuracy issue. This is the ethics connection from the hands-on activity.

Q15 [Type: MCQ | Difficulty: Hard | QID : 53617]

A hospital serving a primarily East Asian population deploys a breast cancer screening AI validated on US and European patient data. Six months in, an internal audit finds the system's sensitivity is 12% lower for their patient population than the published benchmark. What is the most responsible course of action?

- A.** Continue deployment but flag the performance gap in the documentation so radiologists are aware of the limitation
- B.** Keep the system running but lower the flagging threshold to catch more positives overall, accepting the higher false-positive rate
- C.** Limit the system to administrative roles until the vendor releases an updated model trained on more diverse patient data
- D.** Suspend clinical use immediately and validate on a representative patient sample before redeployment

Correct Answer: D

Explanation: Pre-deployment validation on the actual patient population plus ongoing monitoring catches the kinds of performance gaps that have appeared in real CV medical systems. A 12% sensitivity gap means missed cancers; "flag and continue" (A) is insufficient, and lowering thresholds (B) trades false negatives for false positives without fixing the underlying problem.

Q16 [Type: MCQ | Difficulty: Hard | QID : 53619]

A school is considering installing CV systems that can identify individual students through facial recognition. Rank these considerations from most to least important for responsible decision-making.

- I. Privacy rights of students and families, and the chilling effect on student behaviour
- II. Transparency with the school community about what the system does
- III. Whether the school cafeteria's visual aesthetics will be improved
- IV. Accuracy across different skin tones, ages, and appearances, and consequences of misidentification

A. I → IV → II → III

B. IV → III → II → I

C. IV → I → III → II

D. I → II → IV → III

Correct Answer: A

Correct Order: I → IV → II → III

- I. Privacy rights of students and families, and the chilling effect on student behaviour
- IV. Accuracy across different skin tones, ages, and appearances, and consequences of misidentification
- II. Transparency with the school community about what the system does
- III. Whether the school cafeteria's visual aesthetics will be improved

Explanation: Privacy and chilling effects are the foundational ethical questions that drive the deploy/do-not-deploy decision. Demographic accuracy comes next because misidentification has direct consequences for misidentified students. Transparency is necessary but operational - it is about HOW to deploy, not WHETHER. Aesthetics are unrelated to responsibility entirely.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 53620]

Scenario: A CV system trained to recognise cats was only shown training photos of cats sitting on red sofas. When the same cat is photographed sitting on a green sofa, the system becomes confused and fails to identify it.

Question: Which CV challenge is causing this?

A. Lighting

- B. Pose & Angle
- C. Background clutter
- D. Scale

Correct Answer: C

Explanation: The system has accidentally learned the wrong pattern - it tied "cat" to the red sofa rather than to the cat itself. When the background changes, the learned pattern breaks. This is the textbook background clutter problem and the same idea as the "AI that thinks grass means dog" example from training-data variety lessons.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 53621]

Scenario: A smartphone unlocks reliably when the user looks straight at the camera but fails to recognise the same user when their head is turned 90 degrees to the side.

Question: Which CV challenge is at work, and why does it break pattern matching?

- A. Lighting - the side of the face is shadowy, making it harder for the camera to capture features
- B. Pose & Angle - a side face shows different features, so the learned face pattern no longer matches
- C. Bias - the training set excluded this user, so the model never learned to recognise their face
- D. Scale - the face appears too small from the side, so the model cannot detect enough detail

Correct Answer: B

Explanation: A face from the side shows a different set of features - often only one eye, no full nose, no full mouth - than a frontal face. CV recognises by matching combinations of features into a learned pattern; when the arrangement changes, the pattern breaks even though the person is the same.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 53622]

Scenario: You unlock your phone using face recognition.

Question: Walk through the three stages of the CV framework (See → Process → Act) for this everyday action. At the Process stage, name at least one specific feature, one pattern, and the prediction the system would produce.

(Respond in 3–4 sentences.)

Model Answer: In the See stage, the phone's front camera captures the user's face as a grid of millions of pixels with red, green, and blue values. In the Process stage, the system extracts features such as two dark spots in the upper half of the image (eyes), an oval outline (face boundary), and a curved line below the nose (mouth); it combines these into a face pattern - eyes, nose, and mouth arranged in a specific way - and produces a prediction like "this matches the registered user with 95% confidence." In the Act stage, the phone unlocks and shows the

home screen. Throughout, the AI is matching pixel patterns to learned examples, not actually recognising the person the way a friend would.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53623]

Scenario: A city is considering deploying CV systems on traffic cameras that automatically detect drivers running red lights and issue tickets without human review. The transportation director argues this improves road safety and reduces officer workload. Civil rights advocates raise concerns. The council is divided.

Question: Identify one significant benefit AND one significant risk of automated CV-based traffic enforcement, and describe what the city should require to ensure the system is fair and accurate.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that consistent automated enforcement can reduce dangerous driving behaviour more reliably than scattered human enforcement, potentially reducing accidents and protecting drivers, cyclists, and pedestrians. A significant risk is that CV systems can misread licence plates due to lighting, weather, dirt, or unusual fonts, can be biased in accuracy across different camera locations or driving demographics, and removing human review removes the ability to catch system errors before they affect drivers - particularly drivers who may not have the time or means to contest incorrect tickets. To balance these, the city should require human review of every flagged violation before a ticket is issued, transparent reporting of system accuracy broken down by location and demographic patterns, a clear and accessible process for drivers to contest tickets, and an explicit policy that the system can be paused if accuracy problems emerge.